Data Quality Report

AeroGrid | Source: live Neon warehouse | UTC: 2026-11-13T13:41:53.742811+00:00

Audit summary

DQ run: 7; critical FAIL checks: 0; warning checks: 3. Refer to each check below.

Check catalog results

Check ID	Severity	Status	Observed / Expected
DQ-A-001	CRITICAL	PASS	0 / 0
DQ-A-002	CRITICAL	PASS	0 / 0
DQ-A-007	CRITICAL	PASS	0 / 0
DQ-A-008	WARNING	PASS	0 / 0
DQ-B-001	CRITICAL	PASS	0 / 0
DQ-B-003	CRITICAL	PASS	0 / 0
DQ-B-004	CRITICAL	PASS	0 / 0
DQ-C-001	CRITICAL	PASS	0 / 0
DQ-C-003	WARNING	WARN	27479 / 0
DQ-C-005	CRITICAL	PASS	0 / 0
DQ-C-006	CRITICAL	PASS	0 / 0
DQ-C-007	WARNING	PASS	0 / 0
DQ-C-008	WARNING	WARN	27479 / 0
DQ-I-001	CRITICAL	PASS	0 / 0
DQ-I-002	CRITICAL	PASS	0 / 0
DQ-I-003	CRITICAL	PASS	0 / 0
DQ-I-004	CRITICAL	PASS	0 / 0
DQ-I-005	CRITICAL	PASS	0 / 0
DQ-I-006	CRITICAL	PASS	0 / 0
DQ-I-008	CRITICAL	PASS	0 / 0
DQ-I-009	CRITICAL	PASS	0 / 0
DQ-I-010	CRITICAL	PASS	0 / 0
DQ-R-001	CRITICAL	PASS	0 / 0
DQ-R-002	CRITICAL	PASS	0 / 0

Check ID	Severity	Status	Observed / Expected
DQ-R-004	CRITICAL	PASS	0 / 0
DQ-R-005	CRITICAL	PASS	0 / 0
DQ-R-006	CRITICAL	PASS	0 / 0
DQ-T-003	WARNING	WARN	1 / 0
DQ-U-001	CRITICAL	PASS	0 / 0
DQ-U-002	CRITICAL	PASS	0 / 0
DQ-U-003	CRITICAL	PASS	0 / 0
DQ-U-004	CRITICAL	PASS	0 / 0

Real warehouse row counts

Schema.table	Rows
project_staging.stg_adsb_position	27479
project_staging.stg_reject	418
project_staging.stg_airspace_zone	0
project_staging.stg_restriction_snapshot	0
project_dw.dim_time	169
project_dw.dim_site	5
project_dw.dim_vehicle	157
project_dw.dim_spatial_grid	3659
project_dw.dim_weather	1
project_dw.dim_airspace_zone	1
project_dw.fact_vehicle_tracking	27479
project_dw.fact_flight_journey	178
project_dw.fact_close_encounter	0
project_dw.fact_airspace_occupancy	1754
project_dw.fact_restriction_activation	0

Capacity

Actual PostgreSQL database size: 52,781,056 bytes. Object-storage usage is not available from the PostgreSQL connection; do not equate manifest bytes with current bucket usage.

Encounter and restriction limitations

The WP07-v1 encounter algorithm requires identical timestamps. Near-time diagnostics were separate analytical SQL, not the production algorithm. No historical 2022 TFRs are inferred from 2026 FAA services. Zero derived rows may be legitimate source coverage, not evidence of a missing SQL table.

Decision rule

Any stored CRITICAL FAIL blocks trusted publication even if this PDF can be generated. A green unit-test run alone does not establish data quality.

Evidence and reproducibility

Latest ETL run ID: 7, mode=validate, started=2026-11-13 13:19:02.457585+00:00, status=SUCCESS, version=wp08-v1. DQ check IDs map to Phase2/etl/dq.py and Phase2/docs/06_data_quality_and_reports.md. Verify the source manifests and rejected-row counts directly in Neon.